

In [1]: # **1** Project Setup

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# Install Dependencies
#!pip install pandas numpy matplotlib seaborn scikit-learn transformers torch

# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import requests
import re
import nltk
from nltk.corpus import stopwords
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report
from transformers import pipeline
import torch #////////////////////

import os
os.environ["TOKENIZERS_PARALLELISM"] = "false"

# 2 Data Collection & Wrangling

# Fetch News Articles from an API
#API_KEY = "your_newsapi_key_here"
API_KEY="883ccb414ec84e538de73c155d0c2fdf"
URL = f"https://newsapi.org/v2/top-headlines?country=us&category=business&api-key={API_KEY}"

# ----- Start of Troubleshooting Code -----
response = requests.get(URL).json()

# Check the API response
print(response)

# Ensure the 'articles' field exists
if "articles" in response:
    articles = response["articles"]
    df = pd.DataFrame(articles[["title", "description", "content"]])
    df.dropna(inplace=True)
    df["text"] = df["title"] + " " + df["description"]
    df = df[["text"]] # Keeping only relevant column
else:
    print("Error: 'articles' not found in response.")
    print("API Response:", response)

# ----- End of Troubleshooting Code -----

# Text Cleaning
nltk.download("stopwords")
stop_words = set(stopwords.words("english"))

def clean_text(text):
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text = text.lower()
text = re.sub(r"http\S+|www\S+|https\S+", "", text) # Remove URLs
text = re.sub(r"\W", " ", text) # Remove non-word characters
text = " ".join([word for word in text.split() if word not in stop_words])
return text

df["cleaned_text"] = df["text"].apply(clean_text)

# 3 Data Visualization

# Word Count Distribution
df["word_count"] = df["cleaned_text"].apply(lambda x: len(x.split()))
plt.figure(figsize=(8, 5))
sns.histplot(df["word_count"], bins=30, kde=True)
plt.title("Word Count Distribution")
plt.show()

# 4 Sentiment Analysis Using LLMs

sentiment_model = pipeline("sentiment-analysis", model="distilbert/distilbert-base-uncased")
df["sentiment"] = df["cleaned_text"].apply(lambda text: sentiment_model(text).label)

plt.figure(figsize=(6, 4))
sns.countplot(x=df["sentiment"])
plt.title("Sentiment Analysis of News Articles")
plt.show()

# 5 Machine Learning Model for Sentiment Classification

# Prepare Data for ML Model
vectorizer = TfidfVectorizer(max_features=5000)
X = vectorizer.fit_transform(df["cleaned_text"])
y = df["sentiment"].map({"NEGATIVE": 0, "POSITIVE": 1}) # Convert labels to 0 and 1
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Naïve Bayes Classifier
model = MultinomialNB()
model.fit(X_train, y_train)

# Evaluate Model
y_pred = model.predict(X_test)

# Setting zero_division=1 to prevent the UndefinedMetricWarning
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred, zero_division=1))

# 6 Text Summarization Using LLMs
summarizer = pipeline("summarization", model="sshleifer/distilbart-cnn-12-6")

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def dynamic_summarization(text):
    input_length = len(text.split())
    max_length = min(50, input_length + 5) # Adjust max_length based on input
    return summarizer(text, max_length=max_length, min_length=17, do_sample=False)

df["summary"] = df["text"].apply(dynamic_summarization)

print(df[["text", "summary"]].head())

# 7 Deployment (Optional)

# To deploy the model as an API, use FastAPI:
# Install FastAPI and Uvicorn
# !pip install fastapi uvicorn

# FastAPI Code
from fastapi import FastAPI
import uvicorn

app = FastAPI()

@app.post("/predict/")
def predict_sentiment(text: str):
    sentiment = sentiment_model(text)[0]
    return {"sentiment": sentiment["label"], "score": sentiment["score"]}

@app.post("/summarize/")
def summarize_text(text: str):
    summary = summarizer(text, max_length=50, min_length=20, do_sample=False)
    return {"summary": summary}

# Only run uvicorn if not running in Jupyter Notebook
if __name__ == "__main__" and not hasattr(__builtins__, 'get_ipython'):
    uvicorn.run(app, host="0.0.0.0", port=8000)

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nce plunged to the lowest level in five months as concerns over tariffs and how they could raise the cost of housing weighed on the industry.Sentiment fell in Februar... [+114 chars]'], {'source': {'id': None, 'name': 'Ftc.gov'}, 'author': None, 'title': 'FTC Chairman Andrew N. Ferguson Announces that the FTC and DOJ's Joint 2023 Merger Guidelines Are in Effect - Federal Trade Commission News', 'description': 'Federal Trade Commission Chairman Andrew N.', 'url': 'https://www.ftc.gov/news-events/news/press-releases/2025/02/ftc-chairman-andrew-n-ferguson-announces-ftc-doj-s-joint-2023-merger-guidelines-are-effective', 'urlToImage': 'https://www.ftc.gov/themes/custom/ftc_uswds/img/ftc_social_share_default_en.jpg', 'publishedAt': '2025-02-18T15:50:07Z', 'content': 'Federal Trade Commission Chairman Andrew N. Ferguson announced today that the FTC and Department of Justice's joint 2023 Merger Guidelines are in effect and will serve as the framework for the FTC's me... [+399 chars]'}, {'source': {'id': None, 'name': 'Investor's Business Daily'}, 'author': None, 'title': 'European Defense Stocks Rally As Military Worldview Shifts - Investor's Business Daily', 'description': 'European defense stocks rally as developments between President Trump and Russia prompt an emergency meeting. U.S. defense stocks rise.', 'url': 'https://www.investors.com/news/defense-stocks-trump-russia-ukraine-europe-emergency-meeting-lmt-rtx/', 'urlToImage': 'https://www.investors.com/wp-content/uploads/2025/02/BIZ02TrumpPutin021825news.jpg', 'publishedAt': '2025-02-18T15:12:00Z', 'content': 'Information in Investors Business Daily is for informational and educational purposes only and should not be construed as an offer, recommendation, solicitation, or rating to buy or sell securities. ... [+1064 chars]'}, {'source': {'id': None, 'name': 'Investor's Business Daily'}, 'author': None, 'title': 'United Rentals Falls After Losing \$4.8 Billion Deal, Despite Consolation Prize - Investor's Business Daily', 'description': 'United Rentals won't up its \$92 per share cash bid for H&E Equipment Services, which received a higher offer from Herc Holdings. URI stock fell.', 'url': 'https://www.investors.com/news/united-rentals-uri-stock-hees-acquisition-herc/', 'urlToImage': 'https://www.investors.com/wp-content/uploads/2022/12/Stock-UnitedRentals-sign-shutt.jpg', 'publishedAt': '2025-02-18T15:11:00Z', 'content': 'Information in Investors Business Daily is for informational and educational purposes only and should not be construed as an offer, recommendation, solicitation, or rating to buy or sell securities. ... [+1064 chars]'}, {'source': {'id': 'wired', 'name': 'Wired'}, 'author': 'Javier Carbajal', 'title': 'Memecoin Scandal Threatens Argentine President Javier Milei - WIRED', 'description': 'Argentinian president Javier Milei is facing calls for impeachment and a possible criminal investigation for his role in the rise and fast collapse of a memecoin.', 'url': 'https://www.wired.com/story/memecoin-scandal-threatens-argentine-president-javier-milei/', 'urlToImage': 'https://media.wired.com/photos/67b480e5c4c554501e2c7da9/191:100/w_1280,c_limit/milei.jpg', 'publishedAt': '2025-02-18T14:55:38Z', 'content': 'Argentinas president, Javier Milei, has come under fire for the abrupt collapse of a cryptocurrency he promoted, in a move that has triggered lawsuits and calls for his impeachment.\r\nMilei is facing ... [+2999 chars]'}, {'source': {'id': None, 'name': 'Tipranks.com'}, 'author': 'Joshua Enomoto', 'title': 'How Investors Can Defy the Naysayers and Cash In on SMCI Stock - TipRanks', 'description': 'Regarding sheer relevance, it's difficult to overlook the fundamental case for Super Micro Computer (\$SMCI). Specializing in high-performance servers that undergird...', 'url': 'https://www.tipranks.com/news/article/how-investors-can-defy-the-naysayers-and-cash-in-on-smci-stock', 'urlToImage': 'https://blog.tipranks.com/wp-content/uploads/2025/02/shutterstock_1846995598-750x406.jpg', 'publishedAt': '2025-02-18T13:35:29Z', 'content': 'Story HighlightsAlthough AI infrastructure specialist Super Micro Computer got into some hot water, a brewing comeback in SMCI stock may still have some legs left.'}, {'source': {'id': None, 'name': 'Invest

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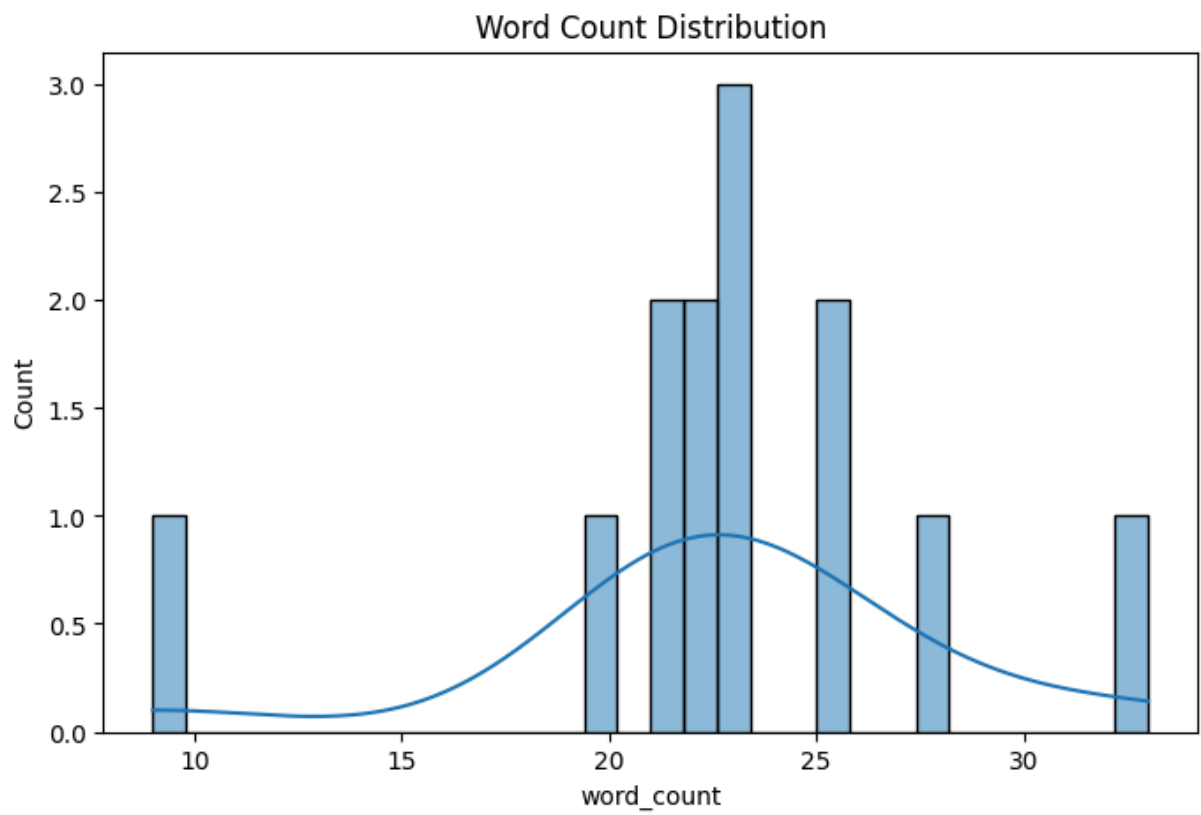
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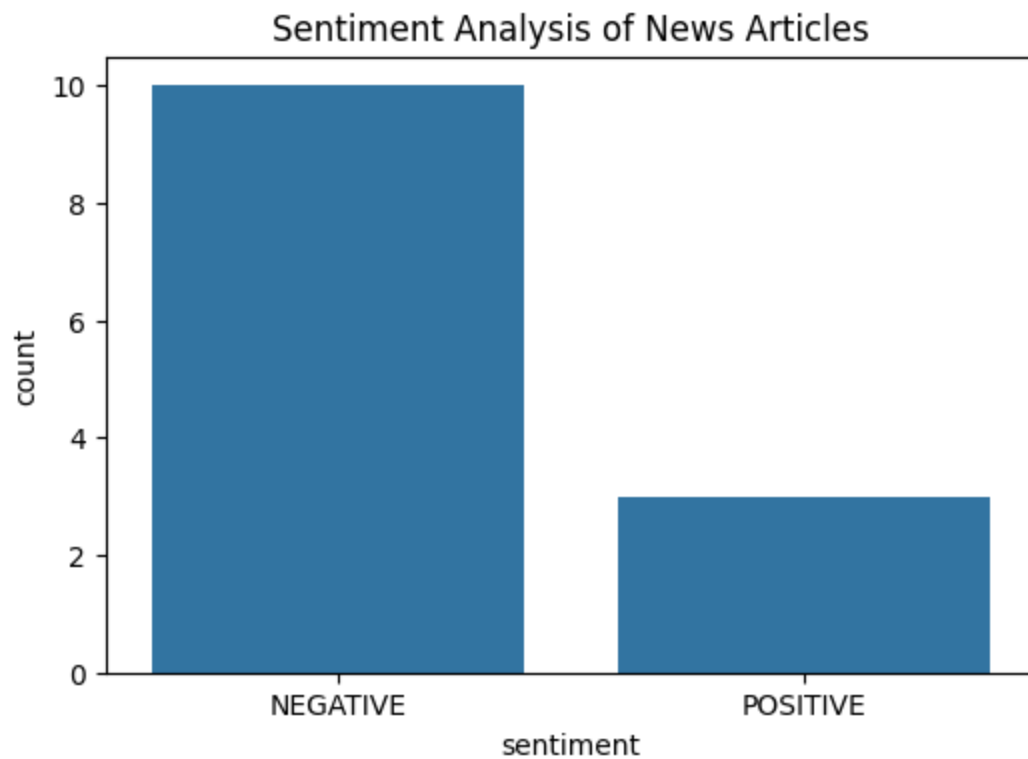
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[nltk_data] Package stopwords is already up-to-date!

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Device set to use cpu



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0	1.00	1.00	1.00	3
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

Device set to use cpu

text \

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7 Home builders just got a reality check – and i...

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summary

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